

Manu Mathew Jiss

San Francisco, Bay Area, USA | [Portfolio](#) | [LinkedIn](#) | [GitHub](#) | [Google Scholar](#)

Education

University of the Pacific, Stockton, CA May 2026
Master of Science in Computer Science

APJ Abdul Kalam Technological University, Kerala, India March 2023
Bachelor of Technology in Computer Science and Engineering

Technical Skills

Programming Languages:	Python, C++, JavaScript, TypeScript, MATLAB, Bash/Shell Scripting, SQL, HTML, CSS
Machine Learning & AI:	TensorFlow, PyTorch, Scikit-learn, YOLOv5, ResNet-50, LLMs, RAG, NLP, Text Classification, Sentiment Analysis, Zero-Shot Learning, Toxicity Detection, BERTweet, HateBERT, DistilRoBERTa, DNN, VADER, TextBlob, SHAP, LIME, Hugging Face Transformers, OpenAI API
Databases & Data Engineering:	MongoDB, PostgreSQL, Redis, ClickHouse, ETL Pipelines, Data Warehousing, Incremental ETL, Metadata Lineage
Backend, Tools & Frameworks:	Flask, FastAPI, Node.js, Express.js, Next.js, Streamlit, REST APIs, Authentication Systems, ROS/ROS2, AWS Lambda, Docker, OpenCV, Jupyter Notebook, VS Code, Git/GitHub, CUDA, JetPack SDK
APIs & Data Sources:	Reddit API, Twitter/X API, YouTube Data API, Yahoo Finance, NewsAPI, Finnhub, Google Cloud APIs, OpenWeather API
Robotics & Hardware:	NVIDIA Jetson, Jetson Orin Nano, LiDAR, ZED Stereo Cameras, IMUs, Sensor Fusion, Raspberry Pi, Arduino, Drones, Embedded Systems
Cybersecurity:	Network Security, Ethical Hacking, Vulnerability Scanning, Penetration Testing, Cryptography, Secure API Integration, Cloud Security Principles
Data & Visualization:	Pandas, NumPy, Matplotlib, Seaborn, Plotly, Chart.js, Recharts, WordCloud, Power BI, Tableau, Excel Analytics, Interactive Dashboards
Systems & Terminal Skills:	Linux (Ubuntu), Embedded Linux, NVIDIA Jetson Orin Nano, JetPack SDK, ROS 2 Humble, colcon, CMake, Docker (ARM64), CUDA setup, SSH/SCP, systemctl, journalctl, OS flashing, system debugging

Startup Experience

- **Co-Founder & Software Engineer** May 2025 – Present
Stock Crusher (GenAI Trading Platform)
 - Built [Stock Crusher](#), a GenAI-powered autonomous trading platform integrating multi-source sentiment (Reddit, news) with LLM-based reasoning to generate real-time BUY/SELL signals and portfolio insights.
- **Co-Founder & Software Engineer** October 2025 – Present
Meal Muse (AI Recipe Assistant)
 - Developed [Meal Muse](#), a full-stack AI recipe assistant with SwiftUI iOS client and FastAPI/Node.js backend, integrating OpenAI-based generation, PostgreSQL caching, and secure authentication (Apple, Google) for personalized recommendations.

Graduate Research Assistant – University of the Pacific

- [Dr. Solomon Berhe](#) May 2025 – May 2026
 - Conducted discourse classification and sentiment trajectory analysis on 542 Reddit software update discussions, distinguishing Technical Problem Solving (TPS) from General Discontent (GDS) using VADER, TF-IDF, and machine learning models.

- Developed author-community sentiment divergence metrics and evaluated NLP classifiers (Naive Bayes, Logistic Regression, RoBERTa); **paper under review at IEEE ICMLA 2026** ([Conference Link](#)).
- *Dr. Don Lee* *August 2025 - December 2025*
 - Developed **SignSight**, a template-guided vision-language framework integrating CLIP semantic reasoning and dual-consistency verification for robust traffic sign recognition under physical adversarial attacks.
 - Achieved 38% higher accuracy than CNN baselines on manipulated traffic signs; **Paper under review at CVPR 2026** ([OpenReview](#)).
- *Dr. Tapadhiri Das* *December 2024 – May 2025*
 - Developed an AI pipeline for sentiment, toxicity, and account authenticity classification using XGBoost and LIME, achieving 96.9% accuracy.
 - Built a risk-profiling framework for bot, scam, spam, and real account detection; **paper under review at IEEE SMC 2026** ([Conference Link](#)).
- *Dr. Tapadhiri Das* *July 2025 – March 2026*
 - Developed LIMFADD, an LLM-generated dataset for Instagram fake account detection, enabling classification into real, spam, scam, and bot categories with 97% accuracy.
 - Designed data collection and XAI-based validation pipeline; **paper published at IEEE ISTAS 2025** ([TechRxiv](#)).
- *Dr. Pramod Gupta* *December 2025 – May 2026*
 - Developed a RoBERTa-based emotion classification and hybrid sentiment analysis framework using the GoEmotions dataset.
 - Built **AIMoodDiary**, an AI-powered journaling platform with mood analytics; **paper under review at IEEE ICMLA 2026** ([Conference Link](#)).

Graduate Teaching Assistant – University of the Pacific

- *Dr. Sethuraman Kuruvimalai* *February 2026 – May 2026*
 - Assisted in teaching a cybersecurity analytics course covering SOC workflows, SIEM, and Splunk, supporting assignment design and evaluation.
 - Guided students in Linux environments, virtual machines, and log analysis, improving hands-on investigation and reporting skills.

Selected Projects & Research

End-to-End Multi-Database Weather Data Engineering Pipeline *November 2025*

- Built an end-to-end weather analytics pipeline for Stockton, CA using Open-Meteo API, MongoDB, ClickHouse, and Redis, implementing incremental ETL with metadata lineage tracking and an Express + Chart.js dashboard for real-time monthly insights ([GitHub Repo](#)).

Pest Detection Using SVM *July 2022*

- Built an SVM-based pest detection system that preprocesses crop images and classifies infected vs. healthy tea leaves with high accuracy, enabling early pest identification for improved agricultural decision-making.

On-Tree Areca Nut Fruit Maturity Detection *December 2022*

- Developed a YOLOv5-based deep learning model integrated with DJI Mini SE drones to detect and classify areca nut maturity in complex backgrounds, optimizing harvest timing and improving yield estimation.